PERFORMANCE EVALUATION OF 3rd ORDER SIGMA-DELTA (Σ-Δ) MODULATORS VIA FPGA IMPLEMENTATION

Saman S. Abeysekera and Charayaphan Charoensak

School of Electrical and Electronic Engineering, Nanyang Technological University, Block S1, Nanyang Avenue, Singapore 639798. E-mail: esabeysekera@ntu.edu.sg, ecchara@ntu.edu.sg

ABSTRACT

Sigma-delta (Σ-Δ) modulators have been widely used over the last few decades, in various signal processing applications. Usually, for improved signal-to-noise (quantization) performances, higher order modulators are utilized in these applications. In this paper, we investigate the performance of third order Sigma-delta (Σ-Δ) modulators and corresponding demodulators via Field Programmable Gate Array (FPGA) implementations. Two modulator architectures, the multi-stage (MASH) Σ-Δ architecture and the Look Ahead Decision Feedback (LADF) Σ-Δ architecture are compared on their performances. It will be shown that the LADF Σ-Δ architecture, which has not been widely reported in the literature, is preferable over the other modulators in the design of Σ-Δ modulators. The selection of demodulator filters for reducing quantization noise resulting from the MASH and LADF Σ-Δ architectures is also addressed in the paper. Advantages and disadvantages of various demodulator filters, e.g. *sinc* filters and *recursive* filters are discussed via the FPGA implementation.

1. INTRODUCTION

For some time, Σ-Δ modulators have been considered for numerous applications in signal processing. These include the conventional A/D and D/A convertor applications as well as the use in the design of FIR and IIR filters, AM/FM modulators, correlators, multipliers and synchronizers [7] [8]. A major reason for the popularity of Σ-Δ architectures lies in their ability to trade bandwidth with quantization noise. These circuits lead to a reduction in expensive hardware and help in speeding up computations besides providing higher resolution in comparison to traditional analogue circuitry. Furthermore, as the Σ-Δ modulated signals are usually restricted to take values of {-1, 0, +1}, the Σ-Δ based signal processing architecture provides programmability and flexibility in the circuits. The 1-bit output of these modulators removes the need for multi-bit multipliers and hence decreases the complexity and cost of the circuit. The resolution is also improved due to over-sampling before modulation. Multipliers can be entirely eliminated from the circuit by using shift registers to process the 1-bit output of the modulator [3]. Though multipliers have become cheaper over the last decade, multi-bit multipliers are expensive for VLSI implementations. Besides, the designed architecture can meet higher bit resolution at lower cost, which makes it very lucrative. All these features makes Σ-Δ based circuit an attractive alternative solution for complete but complicated System-on-Chip designs [7].

This paper presents the FPGA implementation of a rather uncommon LADF sigma- delta (Σ-Δ) modulator [1][8] and an investigation into its hardware architecture. FPGA implementation and comparison of various demodulator filters including a novel (recently proposed in [9]) recursive filter is also discussed. It is demonstrated that LADF presents an attractive architecture for solutions in System-on-Chip circuits.

2. 3rd ORDER Σ-Δ MODULATORS

The original Σ-Δ modulator circuit consists of a single feed-back loop, however, the quantization noise performance of this circuit is usually not adequate for the use in practical applications. The performance of the Σ-Δ modulators can be improved by using multi-loop (identified here as DSM) or multi-stage (MASH) architectures [4]. Recently, another method of Σ-Δ modulation, known as LADF technique, has been proposed by Stonick et. al. [1]. A brief description of the LADF architecture and typical spectral characteristics of above three Σ–Δ modulator architectures are presented in the following sections. The spectral characteristics will be later used for comparison of modulators via FPGA implementations.

2.1 3rd Order LADF (LADF3) Modulator

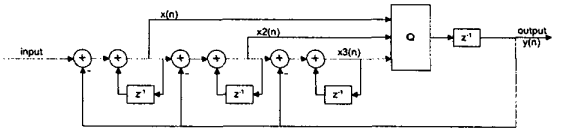

Figure 1: 3rd order LADF modulator

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The multi-loop LADF architecture has two distinguishing features; look-ahead and decision feedback. Thus, the architecture *looks-ahead* at the possible future states and *feeds-back* this information to the current output decision. Figure 1 shows the architecture of a 3rd order LADF modulator, where *x(n), x2(n)* and *x3(n)* comprise the look-ahead section while the feedback section is the return path from *y(n)*. To determine the most stable set of states, the LADF algorithm chooses an output, which minimizes a cost function of the integrators at each time state. The quantizer, which adds the uncertainty or noise to the system can be modeled by an error term. The function *q(n)* of block *Q* in Figure 1 is obtained as follows :

$$q(n) = \mathrm{sgn}\{\frac{1}{6}|x3(n)+1|^3 - \frac{1}{6}|x3(n)-1|^3 + 2(x2(n)+x3(n)) - |x(n)+x2(n)+x3(n)-3| + |x(n)+x2(n)+x3(n)+3|\} \qquad Eq.1$$

where, the *sgn* (or *signum)* function is a 2-level quantizer with output levels {-1,+1} [1].

2.2 Spectral Characteristics of Quantization Noise

The quantization noise power spectrum, $S(f)$, of Σ-Δ modulators can be approximated by the following equations with f denoting the frequency normalized by the over-sampled frequency of the modulator [6][8].

$$DSM2: S(f) = 16\sin^4(\pi.f)/3$$
$$MASH3: S(f) = 32\sin^6(\pi.f)/3$$
$$LADF3: S(f) = 3200\sin^6(\pi.f)/3 \qquad Eq.2$$

A quantity that can be used in the comparison of various Σ-Δ modulator architectures is the filtered quantization noise power resulting from an ideal lowpass filter of bandwidth f_c. Using the results of Equation 2, the filtered quantization noise power - $P(f_c)$, for double loop modulator (DSM2), 3rd order multi-stage (MASH3) modulator and the 3rd order LADF modulator (LADF3) are given by the following equations [8][9].

$$DSM\ 2: P(f_c) = 2f_c - \frac{8}{3\pi}\sin(\pi.f_c) + \frac{1}{3\pi}\sin(2\pi f_c) \cong \pi^4 f_c^5/15$$

$$MASH\ 3: P(f_c) = \frac{5f_c}{16} - \frac{5}{12\pi}\sin(\pi.f_c) + \frac{5}{96\pi}\sin(2\pi f_c) - \frac{1}{6\pi}\sin^4(\pi f/2)_c\sin(\pi.f_c) \cong \pi^6 f_c^7/42$$

$$LADF\ 3: P(f_c) = \frac{1600f_c}{5} - \frac{500}{12\pi}\sin(\pi.f_c) + \frac{500}{96\pi}\sin(2\pi f_c) - \frac{100}{6\pi}\sin^4(\pi f/2)_c\sin(\pi.f_c) \cong \frac{100\pi^6 f_c^7}{42} \qquad Eq.3$$

Note that the approximations in the right-hand side of equation (3) are valid only for small values of f_c. The expressions in equation (3) show that the MASH3 circuit performs better than the other architectures, while LADF3 is better than DSM2. Furthermore, the LADF3 circuit results in *20dB* higher noise power in comparison to the MASH3 architecture. However, the MASH3 circuit output is represented in the integer set {-3,....,+3}, i.e. having seven levels whereas the LADF architecture produces 2-level outputs {-1, +1}. Thus for the same over sampling frequency, the bit rate associated with LADF circuits is less than the bit rate of MASH3 circuit as the former output can be coded using only a single bit while later architecture requires 3 bits to code its output.

3. DEMODULATORS

At the demodulator, the high frequency quantization noise produced by the Σ-Δ modulator is filtered by a lowpass filter to recover the input signal. Suitable demodulator lowpass filters such as Comb filters (also known as Sinc filters), optimum FIR filters [5] and optimum Laguerre (recursive) IIR filters [2] have been proposed in the literature. For the optimum filters and the comb filter, the filtered quantization noise power ψ is given by the following equations:

i) Double loop sigma-delta modulation

Optimum FIR filter	$\psi \approx 60/N^5$
$Sinc^3$ filter	$\psi \approx 243/2N^5$

ii) MASH sigma-delta modulation

Optimum FIR filter	$\psi \approx 8400/N^7$	
$Sinc^4$ filter	$\psi \approx 81920/3N^7$	*Eq.* 4

where N is the length of the filter, which is also the over-sampling ratio. (Note that LADF architecture results in *20dB* higher noise power than a MASH modulator.) Equation (4) shows that the optimum FIR filter is associated with a lower noise power. However, the comb filters have constant coefficients due to which they can be implemented easily, in stages, by using a recursive loop or using shift registers without multipliers. Due to this, comb filters are widely preferred. Simulations have shown that optimum FIR filters as well as comb filters act as whitening filters for quantization noise [8].

3.1 $Sinc^k$ / Comb Filters

An N tap Sinc filter is defined by :

$$h(k) = 1/N \quad (0 \le k \le N-1) \qquad Eq.5$$

An N-tap $sinc^K$ filter is a cascade of K (N/K)-tap filters. Hence the amplitude response of a $Sinc^K$ filter is given by:

$$|H(f)| = \left| K\ \frac{\sin(\pi f N/K)}{N.\sin(\pi f)} \right|^K \qquad Eq.6$$

A linear relationship exists between the order of the modulator used and the number of Sinc filters in cascade. Usually, a $sinc^{L+1}$ filter is chosen to reduce the quantization noise of an Lth order modulator. Note that equation 6 shows

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a gain loss at high frequencies. However, due to the down-sampler in the demodulator, this gain loss is marginal in most modulator applications.

3.2 Optimum FIR and Recursive Filters

Based on the qunatization noise characteristics shown in Equation 2, optimal FIR filters that produce minimum filtered noise at the demodulator have been proposed in the literature. Optimum recursive IIR Laguerre filers have also been proposed for the demodulator in [2]. Although, the optimum filters out-perform the Sinc filters, Sinc filters are attractive from the aspects of efficient hardware implementation. The performance of demodulator filters based on recursive IIR filters has been recently reported in [9]. These recursive filters are optimal and also ideal for FPGA implementation. The filter has a simple architecture suitable for FPGA implementation and could be easily re-designed for a change of the modulator order. Furthermore, the filter bandwidth is also controllable if required [2]. In section 4, the performance of the recursive optimal filter will be discussed.

3.3 Bit-rate Comparison of LADF and MASH

For a bit rate comparison of various Σ-Δ modulator architectures, it is necessary to observe the modulator output coefficient distributions. Following Table 1 shows the coefficient histograms for various modulators, in a typical application. Note that DSM2 and MASH3 modulators were implemented using 3-level quantizers. This is because higher level quantizers result in smaller quantization noise power. However, LADF3 architecture was implemented using a 2-level quantizer.

Coeffs.	-3	-2	-1	0	1	2	3
DSM2	-	-	21	49	30	-	-
MASH 3	2	10	20	30	24	12	2
LADF 3	-	-	45	-	55	-	-

Table 1: Output Coefficient Histograms (%)

The coefficient histograms in Table 1 demonstrate that, although MASH3 architecture is computationally more efficient in noise reduction (refer to Eqn 3), it needs 3-bits to encode its output. In this respect, the LADF architecture is more attractive as its output could be encoded using a single bit. As an example consider CD quality audio applications where it is required to achieve a signal-to-noise ratio of *96dB*. Using Eqn 4 the required bit rate for DSM2, MASH and LADF can be then obtained as 188, 256 and 166, respectively. Increasing the required fidelity further, favors the LADF architecture, e.g. a *120dB* SNR would require a bit rate of 570, 565 and 364, respectively.

4. 3rd Order Σ–Δ Modulator Implementation in FPGA

4.1 Hardware Issues

The 3rd order LADF and the 3rd order MASH modulator were first implemented and tested using Xilinx System Generator™ version 1.0 and MATLAB version 5.3. System generator from Xilinx provides a bit-true and cycle-true simulation of the Simulink™ models under MATLAB environment [10]. The top-level design of MASH3 and LADF3 architectures are shown in figure 2 and 3.

The System Generator was then used to generate VHDL codes from the designs. The VHDL codes were then used to implement the MASH3 and LADF3 modulators using Xilinx Foundation Series™ FPGA development tool. Xilinx Spatan II™ family of FPGA was chosen for synthesis with optimization set for speed. The 3rd order MASH and the 3rd order LADF modulator implementations require 9,200 and 20,200 gates respectively.

Although LADF3 architecture requires more than twice the number of gates when compared to MASH3 design, the price difference of the respective FPGA devices is not significant. The LADF modulated signal contains only two values, -1 and +1 compared to seven values from MASH3 architecture. This simplifies the demodulator design and saves a lot of silicon in the FPGA demodulator devices; i.e. the demodulator design for LADF3 should requires only around one half the number of gates needed for MASH3 implementation.

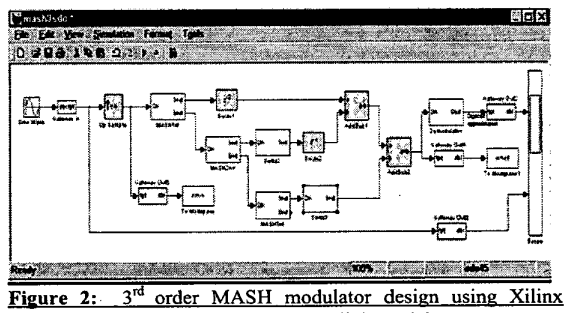

Figure 2: 3rd order MASH modulator design using Xilinx System Generator and MATLAB Simulink model

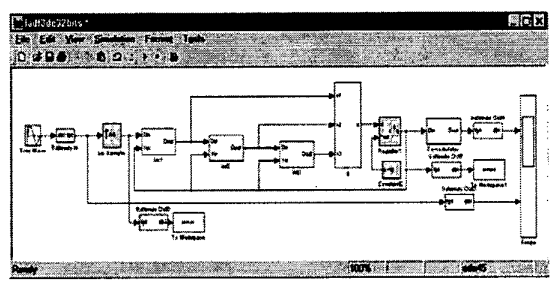

Figure 3: 3rd order LADF modulator design using Xilinx System Generator and MATLAB Simulink model

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4.2 Interpolator / Upsampler

Because of the over-sampling, it is necessary to have an interpolator at the input of modulators. In general, ideal interpolation needs expensive multipliers in the design. As an alternative interpolation technique, the signal can be copied *(N-1)* times in between the input samples. This method is simple yet effective and does not employ multipliers in the hardware design. Furthermore, because of the down-sampling operation, the high frequency roll-off effect of such interpolation has only a marginal effect on the output.

4.3 Performance Results

This section shows modulator output results, obtained from FPGA implementations of LADF and MASH architectures. Figure 4 and 5 depict a typical output from LADF modulator and the corresponding accumulated output noise power, $P(f_c)$. Figures 6 and 7 are similar plots but for MASH architecture. The results of Figure 5 and 7 justify the use of approximate spectral characteristics described in equation 3.

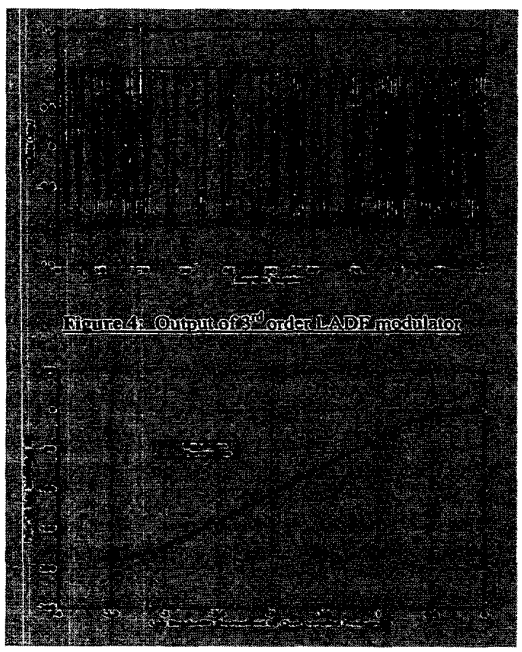

Figure 5: Accumulated Output Noise Spectrum, $P(f_c)$, Resulting from 3rd order LADF modulator

5. De-Modulator Implementation in FPGA

5.1 Hardware Issues

The sinc filter [2] and optimum recursive filter [9] were first designed using System Generator. The generated VHDL codes from System Generator were then synthesized using Xilinx Foundation development tools. The Simulink model was set to optimize the mapping of Xilinx CORE Generator™ to create hardware macros. This ensures accurate hardware implementation when the generated VHDL codes are synthesized. The inc filter and the optimum recursive filter implementations, using the same optimization setting, require 41,100 and 37,200 gates respectively.

Although the total number of gates saved is marginal, many advantages of this implementation, such as low number of multipliers and delays, make this architecture especially attractive for hardware implementation. This is especially true for high-speed applications [9].

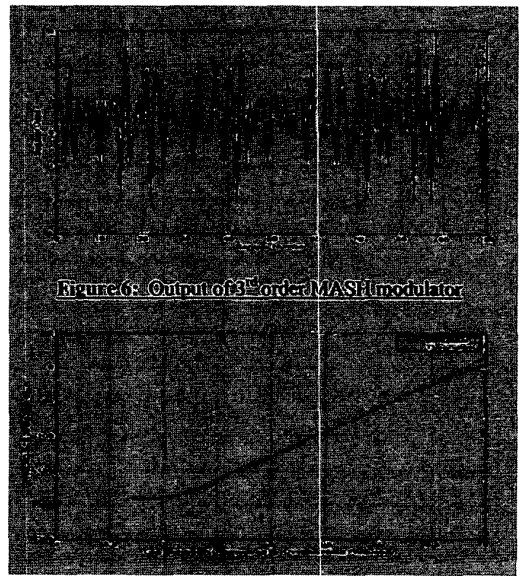

Figure 7: Accumulated Output Noise Spectrum, $P(f_c)$, Resulting from 3rd order MASH modulator

5.2 Performance Results

This section presents the results obtained from FPGA implementations of de-modulators for LADF and MASH architecture. Both sinc filters and optimum recursive filters have been implemented and their performances are almost identical. Results of Figure 8 and 9 are obtained from using 3 sinusoids as the input (amplitudes and normalized frequencies respectively are: 0.36, 0.24, 0.18 and 0.03, 0.13, 0.35). The figures depict the demodulated input and output signals and their spectra resulting from LADF architecture. (The measured quantization noise level for this input is *–60dB* whereas the theoretical value form Eqn 4 is given by *–67dB*.) Figure 10 and 11 depict the input (single sinusoid of amplitude 0.5 and normalized frequency 0.0625) and the demodulated output signal and their spectra resulting from MASH architecture. (The measured quantization noise level for this input is *–88dB* whereas the theoretical value form Eqn 4 is given by *–87dB*.)

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6. SUMMARY

Sigma-delta modulators provide high resolution as compared to analogue circuits and could be effectively used to implement various signal processing algorithms on a System-on-Chip basis. Third order Σ–Δ modulator architectures, LADF and MASH have been compared in this paper by considering FPGA implementation issues. It has been found that, in general, these higher order modulators perform better in terms of noise reduction and lower oversampling ratio requirements. MASH architecture provides lower quantization noise power than LADF architecture. However, because of its single-bit output format, LADF has simplicity in hardware architecture and, therefore, is an appropriate choice for FPGA based SoC implementations. The performance and implementation of corresponding demodulator circuits have also been discussed in the paper. It has been shown that an optimal filter designed via a recursive filter approach presents an attractive FPGA solution for the implementation of the demodulator.

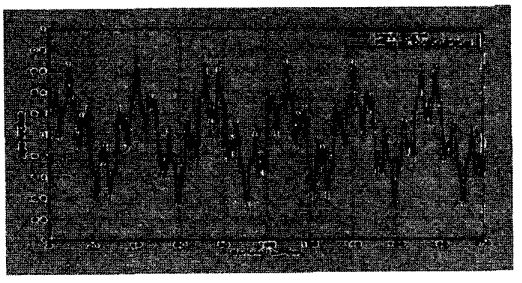

Figure 8: Input (3 sinusoids) and the Demodulated Output of 3rd order LADF modulator

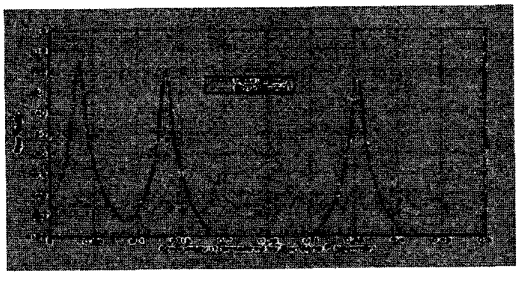

Figure 9: Input (3 sinusoids) and the Demodulated Output Signal Spectra of 3rd order LADF modulator

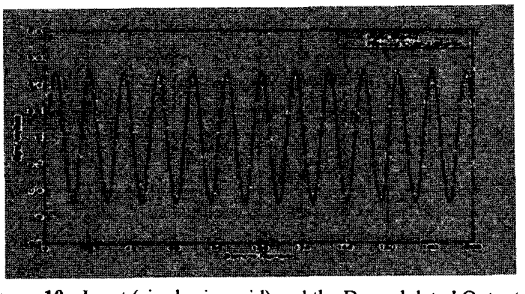

Figure 10: Input (single sinusoid) and the Demodulated Output of 3rd order MASH modulator

Figure 11: Input (single sinusoid) and the Demodulated Output Signal Spectra of 3rd order MASH modulator

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